

A/B Test Example

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Statistical Inference

Universe

- Frequentist: $\{z_statistic, p_value, ci_diff, cohens_h, is_significant\}$
- Bayesian: $\{prob_b_better, expected_lift, credible_interval_lift\}$
- Bootstrap: $\{ci_diff\}$
- Power: $\{observed, n_for_80pct\}$

Variables

Symbol	Meaning	Type / Range
μ_1, μ_2	population means for groups 1 and 2	$\mathbb{R}$
$\bar{x}_1, \bar{x}_2$	sample means	$\mathbb{R}$
s_1, s_2	sample standard deviations	$\mathbb{R}_{>0}$
n_1, n_2	sample sizes	$\mathbb{Z}^+$
t	test statistic (hypothesis test)	$\mathbb{R}$
p	p-value	$[0, 1]$
α	significance level	$[0, 1]$

Structure

- **Control (A)**: 1,500 visitors, 60 conversions (4.00%)
- **Treatment (B)**: 1,500 visitors, 85 conversions (5.67%)
- **Question**: Is the new checkout significantly better?

Real-World Mapping

Real-World Concept	Mathematical Object
group average	sample mean $\bar{x}$
group variability	sample std s
sample count	n
significance threshold	$\alpha = 0.05$

Constraints

(1)

$$H_0 : \mu_1 = \mu_2$$

null hypothesis

(2)

$$H_1 : \mu_1 \neq \mu_2$$

alternative hypothesis

(3)

$$\alpha = 0.05$$

significance level

(4)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Welch's t-test statistic

Objective

Reject H_0 if $p < \alpha$

Approach

Suggested approach Two-proportion z-test + Bayesian Beta-Binomial + Bootstrap

Complexity class:

Available tools: Two-proportion z-test + Bayesian Beta-Binomial + Bootstrap

2. Solution

Answer:	Statistical test complete
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	Two-proportion z-test + Bayesian Beta-Binomial + Bootstrap
Time:	0.2076s
Certificate:	Assumptions checked; test statistic independently verified.

Solution Details

Bayesian Analysis:

Power Analysis:

Recommendation: CAUTIOUS: Statistically significant but small effect. Consider practical impact.

Verification

✓ Rate A Correct	Passed
✓ Rate B Correct	Passed
✓ Bayesian Direction Agrees	Passed
✓ All Methods Agree	Passed

3. Interpretation

The Question

- **Control (A)**: 1,500 visitors, 60 conversions (4.

The Answer

Test complete.

What This Means

- Z-statistic: ~2.36, p-value: ~0.018 (significant at alpha=0.05)
- Cohen's h: ~0.130 (small effect)
- P(B > A): ~98.9% (Bayesian)
- Power: ~65% (underpowered for this effect size)
- Recommendation: "CAUTIOUS" -- significant but small effect

Sensitivity Analysis

Parameter	Current	Change	New Obj.	Class
double_sample_size	present	removed		robust
half_effect	present	removed		robust
power_curve	present	removed		robust

Recommendations

1. Report effect sizes alongside p-values for practical significance assessment.

Limitations

- Results are conditional on model assumptions (normality, independence, etc.).

• Statist